

Practice Quiz 03B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 03

How the universe of discourse affects a quantified statement.

1. Determine whether each statement is true or false. For a true statement, give a formula for y in terms of x . For a false statement, give a specific x for which no allowed y works and explain why.

(a) $\forall x \in \mathbf{R} \exists y \in \mathbf{R} (2y = x)$.

Solution This statement is true. For any real x , take $y = x/2$. This is a real number and satisfies $2y = x$. ■

(b) $\forall x \in \mathbf{Z} \exists y \in \mathbf{Z} (2y = x)$.

Solution This statement is false. Take $x = 1$. The only real solution of $2y = 1$ is $y = 1/2$, which is not an integer, so there is no allowed y . ■